RAG Intelligence System

Lylah Murrah, Isaac Guttormson

What the System does:

It retrieves passages most likely to contain an answer from a question then generates an answer grounded in those passages, and shows which documents the answer came from.

Who Built What:

Lylah: Test Questions, Evaluation Set, Gold Questions, Pipeline, Failure Taxonomy, Evaluation Metrics

Isaac: Documents, Chunking, Corpus, Ingestion Plan, Readme, Executive Summary, Results

User

Our RAG Intelligence System is made for Beloit College students that need information contained in Beloit College's Course Catalog and Student Handbook.

Our system should be able to answer any general questions that have answers contained in these documents like:

Does Beloit College have an Engineering Degree?

How many units is a Greek and Latin Studies Major?

When does summer term begin in 2027?

This system is an alternative to manually finding answers in the documents, which can be difficult to find

Corpus

Our corpus is Beloit College’s Course Catalog and Student Handbook

The Course Catalog is 225 pages and the Student Handbook is 122 creating 143 documents and 369 chunks

The only pages left out were pages with no information (copyright or images)

Splitting

Each document contains an index, each labeled section in the index was used for an initial split.

Occasionally, multiple sections were contained in one page and that one page was uploaded multiple times.

An example of a wrong split would be the academic calendar for 2026 and 2027. We thought it was only one page like the 2025 one so we needed to go back and fix it.

Chunking

We used an initial chunk size of 750 with no overlap

Embedding model:

```
embedder = Pipeline([
    ("features", FeatureUnion([
        ("words", TfidfVectorizer(ngram_range=(1, 2), stop_words="english",
                                  sublinear_tf=True, strip_accents="unicode")),
        ("chars", TfidfVectorizer(analyzer="char_wb", ngram_range=(3, 5),
                                  sublinear_tf=True, min_df=2, strip_accents="unicode")),
    ])),
    ("unit_length", Normalizer()),
])
```

The default goes by every word but by using our settings going by one or two we allow to get combinations like course codes, two word concepts, and proper names.

Retrieval

Chunks were retrieved by a similarity score to the question, the highest scoring chunks were used for answering

If no chunks had a high enough score, the model would refuse to answer the question

Answer Step

From the retrieved chunk, the model searches for an answer, looking for a specific quote that can give a solution

If there is no part of the chunk that answers the question, the model refuses to answer the question

```
schema = types.Schema(
    type=types.Type.OBJECT,
    properties={
        "answer_found": types.Schema(
            type=types.Type.BOOLEAN,
            description="True if context contains the answer, False if answer is missing."
        ),
        "answer": types.Schema(
            type=types.Type.STRING,
            description="The answer to the user's question, or a refusal if answer is missing."
        ),
        "supporting_quote": types.Schema(
            type=types.Type.STRING,
            description="The exact quote from the context that contains the answer. Empty if answer is missing."
        ),
        "source_doc": types.Schema(
            type=types.Type.STRING,
            description="The source filename that the quote came from. Empty if answer is missing."
```

Metrics

Hit rate : How many answerable questions had the expected document in the top three chunks it pulled from. Does not measure the quality or relevance of the content within those chunks beyond their presence

Quote fidelity: When the system answered the question was the answer actually in the chunks it pulled from. Does not measure whether the answer provided is factually correct

Correctness: Do the answers given by the system match the predetermined answers.Does not measure the clarity, conciseness, or safety of the generated response

Correct refusals: Did the system refuse the ones that were unanswerable. Does not measure why the question was unanswerable

Wrong refusals: Did the system refuse to answer questions that were answerable. Does not measure why the system failed to answer an answerable question

Baseline

Chunking by 750 words

Hit Rate: 23 of 26

Quote Fidelity: 19 of 19

Correctness: 23 of 30

Correct Refusals: 4 of 4

Wrong Refusals: 7 of 26

Changes

Changing chunk size to 450:

Hit Rate: 24 of 26

Quote Fidelity: 19 of 20

Correctness: 24 of 30

Correct Refusals: 4 of 4

Wrong Refusals: 6 of 26

Changes

Changing the chunk size to 150:

Hit Rate: 22 of 26

Quote Fidelity: 18 of 18

Correctness: 20 of 30

Correct Refusals: 4 of 4

Wrong Refusals: 8 of 26

Results

Chunk size of 450 gave the best result in the testing dataset

Chunk size of 150 gave the worst results

Both 750 and 450 had very similar results overall for most questions

What It Costs

Overall the similarity scores remained somewhat consistent across all chunk sizes

3 questions had different answers between chunking by 750 and 450, but chunking by 450 was only one question better

When testing chunking by 150 it gave the worst result but was able to answer two question the other 2 couldn’t, listing faculty of departments

Evaluation Set

How we wrote the gold questions: Read through the Course Catalog and Handbook picking questions from sections that seemed interesting and a student would ask.

What counted as expected evidence: The section that contains the answer that we wanted out of the model.

26 answerable questions mixing in different phrasing on how to ask the questions and 4 non answerable questions

One Verified Answer

Question: Can a dog be an ESA?
Answer: Yes, a dog can be an Emotional Support Animal (ESA)

[1] Student Handbook Policies and Regulations F1 | pdf pages 56-65 | hb_section3_f1__02
An ESA can be a dog, cat, small bird, rabbit, hamster, gerbil, fish, turtle or other small, domesticated animal that is traditionally kept in the home. A dog must be at least 9 months of age to live on campus to assure that the dog is reliably housebroken, not disruptive to other residents, and has all of the shots required by state and local ordinance as necessary to make it safe to be around humans. Other types of ESAs must

[2] Student Handbook Policies and Regulations F2 | pdf pages 65-66 | hb_section3_f2__01
The student, not the College, is responsible for the actions of the Service Dog including bodily injury or property damage. Students with Service Dogs are likely to be charged if additional cleaning or damage occurs as a result of having the Service Dog on campus. The student is expected to pay these costs upon repair or cleaning. In addition, the College retains the right to remove the Service Dog, at the owner's expense, should the Service Dog become a direct threat to the health and safety of others or violate these requirements in any way.

[3] Student Handbook Policies and Regulations F1 | pdf pages 56-65 | hb_section3_f1__06
The student, not the College, is responsible for the actions of the Service Dog including bodily injury or property damage. Students with Service Dogs are likely to be charged if additional cleaning or damage occurs as a result of having the Service Dog on campus. The student is expected to pay these costs upon repair or cleaning. In addition, the College retains the right to remove the Service Dog, at the owner's expense, should the Service Dog become a direct threat to the health and safety of others or violate these requirements in any way.

Where It Fails

Specifically questions that are only referenced in a few lines, primarily in the course catalog, “Who are the Critical Identity Studies Faculty?” for example

Listing specific and individual faculty was not "Who is the Community Connections Faculty?" and “Who are the Morse Library Staff” was answered correctly through all 3 chunk sizes

What We Did

This was a reason for why we tested lower chunk sizes, chunking by 150 was able to answer the general faculty questions, at the expense of answering many of the other questions incorrectly

Rewording the questions to make it more specific was also done, though we could not find a wording that changed the output

Ship It

Would you let a first-year student use this unsupervised?

No, only returns a relevant answer about 80% of the time, not enough to be used in place of manually finding the answer. This model can be used to help find information on specific topics using its references to sections of the documents, it almost always gives relevant sections

Limitations

Double Counting Sections - Parts of the documents had multiple sections on one page, these were double counted in our chunking process, cluttering the similarity scores

Answer Complexity - Most of the answers are still simple, primarily only giving a yes or no answer with a somewhat related quote. A more specific answer would be better